WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-08-03 to 2026-08-09

Curated articles in political communication and computational social science.

SUMMARY

This week's COMAP Journal Bot brief brings together a particularly strong set of work on how people encounter, interpret, and act on information. New studies trace political knowledge gaps in online search, gendered issue publics on Twitter, immersion through point-of-view editing, early social media use and school outcomes, and the individual harms produced by redistricting.

Three themes stand out for the lab:

- Digital environments distribute both opportunity and disadvantage: access to search, social platforms, and AI does not translate automatically into equal learning or participation.
- Multimodal and computational approaches are becoming more closely tied to substantive communication theory, from neural responses to audiovisual editing to higher-order network structure.
- Institutional design matters across domains, including redistricting, war escalation, evidence-based policy, climate governance, immigration enforcement, and AI adoption in education.

This digest was prepared early on Saturday, 2026-08-08, before the nominal window closes on Sunday, 2026-08-09. It reflects records available at collection time; publications appearing later in the window may be recovered in a subsequent catch-up or late-additions section.

HIGHLIGHTS

The Knowledge Gap in a High-Choice Media Environment: Experimental Evidence from Online Search

POLITICAL COMMUNICATION · 2026-08-06

Why it matters · Combining randomized encouragement with passive browser tracking, the study shows why equalizing search activity alone may not close political knowledge gaps.

Open article →

Through their eyes: neural and behavioral evidence for enhanced immersion via point-of-view editing

JOURNAL OF COMMUNICATION · 2026-08-04

Why it matters · The article links a concrete editing technique to shared attention, memory, and neural signatures of perspective-taking in audiovisual narratives.

Open article →

Twitter Conversations: Gendered Issue Publics Among Women

COMMUNICATION RESEARCH · 2026-08-03

Why it matters · Its elite and non-elite comparison clarifies when gendered political issue publics emerge online-and where expected differences do not appear.

Open article →

Individual and differential harm in redistricting

NATURE HUMAN BEHAVIOUR · 2026-08-06

Why it matters · A counterfactual, voter-level harm framework extends redistricting evaluation beyond partisan bias to racial, class, geographic, and local contexts.

Open article →

Collective problem decomposition improves the wisdom of deliberative crowds

NATURE COMMUNICATIONS · 2026-08-05

Why it matters · Across online deliberations and experiments, collective decomposition emerges as a measurable mechanism for improving group judgment.

Open article →

Escalation dynamics and the severity of wars

SCIENCE ADVANCES · 2026-08-07

Why it matters · Within-conflict data show that escalation-not simply duration or a violent start-is the main mechanism producing exceptionally severe wars.

Open article →

Longitudinal effect of early social media use on standardized learning outcomes during school career

NATURE HUMAN BEHAVIOUR · 2026-08-03

Why it matters · Longitudinal difference-in-differences evidence connects earlier account adoption to lower standardized learning outcomes and identifies smartphone pervasiveness as a mediator.

Open article →

Higher-order trade-offs in hypergraph community detection

SCIENCE ADVANCES · 2026-08-07

Why it matters · The paper makes model-selection trade-offs explicit for higher-order networks, offering useful foundations for computational studies of complex social structure.

Open article →

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Previous Week Catch-Up

- Political Communication
- Communication Research
- Political Analysis
- American Political Science Review
- Nature Communications
- Proceedings of the National Academy of Sciences
- Science Advances
- Nature
- Science

COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
289	161	128
LATE ADDITIONS CANDIDATES	**CURATED DIGEST BELOW**	**NOTE**
0	23 new, 23 catch-up, 0 late additions	These counts reflect the complete collected set before manual relevance filtering, while the attached PDF reflects the curated digest.

FULL CURATED DIGEST

New This Week

Political Communication

The Knowledge Gap in a High-Choice Media Environment: Experimental Evidence from Online Search

2026-08-06 · DOI 10.1080/10584609.2026.2708131

Roberto Ulloa; Leonard Tiedemann; Peter Selb; Celina Kacperski

Humboldt University of Berlin, Germany; Seeburg Castle University, Austria; University of Konstanz

Abstract. Persistent inequalities in political knowledge are a central concern in political communication. We organize the mechanisms underlying the knowledge-gap literature by distinguishing between individual preconditions, structural features of the information environment, and topic characteristics. Within this framework, we note that self-directed information seeking, a prototypical form of intentional exposure, has received little attention despite its importance in navigating today's complex information environment. We conducted a field experiment in Germany combining randomized encouragements and passive browser tracking to examine how individuals with varying education levels acquire policy-specific knowledge through online search. Participants were randomly assigned to one of three conditions (verbal encouragement, financial encouragement, or control) to seek information on three salient policy topics differing in divisiveness and complexity (child support, energy transition, and cannabis legalization). We estimate both intention-to-treat (ITT) and local average treatment effects (LATE) of information seeking on post-search knowledge outcomes, with a focus on education and civic knowledge as moderators. While the interventions equalized information-seeking behavior, the results provide some support for the knowledge gap hypothesis: knowledge gains were concentrated among participants with higher education or baseline civic knowledge, who, according to our post-hoc exploratory analyses, appeared more effective at navigating search results. These findings indicate that a narrowing of knowledge inequalities goes beyond motivation: it calls for both individual-level interventions to strengthen citizens'skills and structural-level adaptations to foster more equitable learning environments.

Read article →

Journal of Communication

"Communicating in the Air Force": the National Society for the Study of Communication, the Journal of Communication , and U.S. military education, 1950-1962

2026-08-05 · DOI 10.1093/JOC/JQAG012

Dominique Trudel; David W Park

Department of Communication, Lake Forest College, Lake Forest, USA; Ecole des medias, Universite du Quebec a Montreal, Montreal, Canada

Abstract. Abstract This article explores the intertwined history of the National Society for the Study of Communication (NSSC), its Journal of Communication, and U.S. Air Force educational programs. Founded in 1950 (and renamed the International Communication Association in 1969), NSSC developed a close partnership with the Air Force's Air University in its early days. The growth of the academic society and the expansion of communication courses in military education programs were then mutually reinforcing: NSSC leaders-including several Air Force officers and civilian consultants-helped define and refine military communication curricula. Air University resources supported NSSC projects, most notably the Journal of Communication. Drawing on extensive archival research and a close reading of the Journal's content during its first decade, this article argues that these collaborations not only nurtured NSSC's institutional development, but also shaped a

specific conception of communication emphasizing vocational training, efficiency, and the transmission of command.
Read article →

Through their eyes: neural and behavioral evidence for enhanced immersion via point-of-view editing

2026-08-04 · DOI 10.1093/JOC/JQAG011

Moran Ovadia; Shiri Davidovitch; Romi Sion; Naama Shoham; Nimrod Keren; Sergei Freedman; Cynthia Cabanas; Dov Rubinstein; Mark A Thornton; Jonathan Frome; Liad Mudrik; Tim J Smith; Gal Raz

Creative Computing Institute, University of the Arts London , London,; Department of Communication, Vrije Universiteit Amsterdam , Amsterdam,; Department of Psychological and Brain Sciences, Dartmouth College , Hanover, NH,; National University of Singapore NUS College, , Singapore,; Sagol Brain Institute, Tel Aviv Sourasky Medical Center , Tel-Aviv,; Sagol School of Neuroscience, Tel Aviv University , Tel-Aviv,; School of Psychological Sciences, Tel Aviv University , Tel-Aviv,; Steve Tisch School of Film and Television, Faculty of the Arts, Tel Aviv University , Tel-Aviv,

Abstract. Abstract Point-of-view (POV) editing-the juxtaposition of a character's gaze with the object of that gaze-is presumed to function as a unique empathic device that dissolves viewer-character boundaries; however, direct empirical evidence remains scarce. This study provides the first systematic functional magnetic resonance imaging (fMRI) and behavioral investigation into whether POV editing facilitates shared attention and intentionality. 39 adults viewed 8 original narrative movies containing POV sequences or closely matched control conditions. POV sequences increased fixation durations on characters' faces and enhanced subsequent recognition of gazed-at content. Neurally, POV editing elicited stronger responses in regions implicated in mentalizing and increased shared, item-specific activation patterns across viewers within areas associated with perspective-taking and goal-directed control. Multivariate decoding further indicates that a character's gaze induced more robust neural signatures of character identity on subsequent target elements. Together, these findings support a functional role for POV editing in promoting joint attention and shared intentionality in audiovisual messages.
Read article →

Communication Research

Twitter Conversations: Gendered Issue Publics Among Women

2026-08-03 · DOI 10.1177/00936502261470086

Caley Hewitt; Ayla Oden; Nichole M. Bauer

Louisiana State University, Baton Rouge, USA

Abstract. We investigate how gender shapes women's political discussion on Twitter using an issue publics framework. We test whether the conditions for gendered issue publics to form among women around feminine issues exist using data on political discussion via social media in the 2020 and 2022 elections. Our data includes political discussion among political candidates, elites, and non-elites, or regular people. Our results reveal several key findings. First, we find that men elites were more likely to discuss feminine issues, such as health care, compared to women elites, but we found no gender differences between regular Twitter users in our sample. Democratic women elites are more likely to discuss equity-based feminine issues, such as equal pay, than Republican women elites, but we find no partisan differences among non-elite women. We also find no gender or partisan differences at the elite and non-elite levels in the discussion of trait-based feminine issues, such as education. Finally, we show that in 2022, the Dobbs decision that rescinded the constitutional right to an abortion in the U.S. increased the salience of this issue among both women and men. Overall, our research suggests that the conditions for issue publics among women may not strongly exist as we find that non-elite men talk about feminine issues too.
Read article →

Political Analysis

Let Them Eat Pie: Addressing Sample Selection in Multiparty Elections

2026-08-05 · DOI 10.1017/PAN.2026.10049

Ali Kagalwala; Thiago M. Q. Moreira; Guy D. Whitten; Yongzhi Xu

Louisiana State University; Syracuse University; Texas A&M University; University of International Business and Economics

Abstract. Abstract Elections are central to the study of politics. When studying parties' vote shares across districts, scholars are encouraged to use compositional-outcome models in order to test their theories about what factors shape the dynamics of electoral support. Existing compositional modeling approaches incorrectly deal with scenarios in which not all parties compete in every electoral district. Because unobserved factors that affect a party's decisions to contest a district are likely correlated with its performance in districts where the party fielded candidates, failing to account for partial contestation is likely to result in sample selection bias. Addressing sample selection in a compositional setting is challenging because the outcomes are in log-ratio form, and thus the errors often deviate from normality. To deal with these issues, we introduce a novel maximum likelihood approach which accounts for this type of sample selection and demonstrate through simulations that our method outperforms commonly used solutions, including the conventional Heckman correction. We illustrate the utility of our approach by analyzing the 2017 and 2019 U.K. parliamentary elections in English constituencies.

Read article →

Nature Human Behaviour

Individual and differential harm in redistricting

2026-08-06 · DOI 10.1038/S41562-026-02511-7

Cory McCartan; Christopher T. Kenny

Abstract. Social scientists have developed dozens of measures for assessing partisan bias in redistricting. However, these measures are not easily adapted to other groups, including groups defined by race, class or geography, nor are they applicable to single- or no-party contexts, such as local redistricting. Here we propose a unified framework of harm for evaluating the impacts of a districting plan on individual voters and the groups to which they belong, to overcome these limitations. We consider a voter harmed if their chosen candidate is not elected under the current plan but would be under a different plan. Harm improves on existing measures by both focusing on the choices of individual voters and directly incorporating counterfactual plans. We discuss strategies for estimating harm using redistricting simulations, and demonstrate the utility of our framework through three applications to US redistricting. Overall, harm provides a flexible way to precisely quantify redistricting's individual impacts. McCartan and Kenny develop a framework for quantifying the individual-level impacts of redistricting in the USA, showing its applicability for identifying partisan gerrymandering and differential effects across populations.

Read article →

Longitudinal effect of early social media use on standardized learning outcomes during school career

2026-08-03 · DOI 10.1038/S41562-026-02522-4

Marco Gui; Chiara Respi; Giovanni Abbiati; Cristiana Paladini; Sofia Ercolanoni; Francesca Milzani; Tiziana Pirola; Giovanni Vezzoli

Abstract. Abstract This study uses difference-in-differences models to evaluate the impact of early social media use on academic performance among children aged 7-16 years. A dataset comprising 5,227 students from northern Italy was constructed by merging data from a questionnaire on lifetime digital media habits with longitudinal standardized test scores in Italian, English and mathematics. Participants who created a personal social media account in 6th, 7th or 8th grade were

separately matched with late adopters who initiated their first social media account in the 9th grade or later. The findings provide compelling evidence that early engagement with social media adversely affects learning outcomes. Additional findings on the effects of early social media use on other academic outcomes for which we do not have longitudinal data are in line with our main findings. We also find supporting evidence that a substantial part of this negative effect is mediated by students' smartphone pervasiveness in key moments of the day.
Read article →

Nature Communications

Human learning of probability distributions is biased toward moderate structural complexity

2026-08-08 · DOI 10.1038/S41467-026-76247-3

Tianyuan Teng; Li Kevin Wenliang; Hang Zhang

Abstract. Abstract Inferring hidden environmental structures, which commonly involves learning arbitrary probability distributions from limited samples, is essential to optimal and adaptive behaviors across various cognitive domains. However, it remains largely unknown how the internal representations constructed by humans may deviate from actual probabilistic structures, and what computational processes, operated under inherent cognitive limitations, give rise to these representations. We first develop a structured distribution report task to reveal human participants' internal representations, with findings verified in a further distribution recognition task. Across eight behavioral experiments (including one pre-registered study) in two modalities, participants estimate the overall probability density reasonably well, but exhibit a systematic bias toward moderate structural complexity, reporting too many clusters when the true distribution is a single Gaussian and too few when it contains many clusters. We then build a series of learning models in the framework of approximate Bayesian inference fit to our behavioral data. Through model comparisons, we reconstruct the prior beliefs guiding the evolution of participants' internal representations. The best-fitting model for human reports reduces structure growth rate as complexity increases, effectively constraining the complexity of internal representations within memory limitations.
Read article →

Strategic timing for the decarbonization contributions by China's photovoltaic manufacturers in global energy transition

2026-08-06 · DOI 10.1038/S41467-026-76424-4

Qingyang Lin; Chenglin Li; Kai Wang; Kai Fang; Meixuan Xin; Siqi Wang; Anqi Xu; Zhenci Xu; Yue Lu; Xiang Gao

Abstract. Abstract Photovoltaic (PV) technology is central to clean electricity and the Sustainable Development Goals. However, China's PV manufacturing industry, which accounts for nearly two-thirds of global capacity, is often criticized for its high carbon emissions. Existing studies typically assign emission responsibility to manufacturers while attributing decarbonization benefits to power generation, overlooking upstream producers' contributions. Here we propose a Producer Decarbonization Contribution framework to quantify these long-overlooked contributions at national and global scales. We show that China's PV manufacturers contribute 31.4%-73.8% of total PV decarbonization in China and 27.9%-45.4% globally. We further identify a strategic timing, defined as the point at which the PV's decarbonization contributions reach their maximum, occurring around 2035 in China under the Announced Pledges Scenario and 2030 globally under the Net Zero Emissions Scenario proposed by the International Energy Agency. These findings suggest maintaining supportive policies for China before 2035, while adopting more stringent measures globally after 2030 to avoid overproduction. The proposed framework that enables adaptive industrial policy design can be extended to other renewable energy technologies.
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Collective problem decomposition improves the wisdom of deliberative crowds

2026-08-05 · DOI 10.1038/S41467-026-76365-Y

Federico Barrera-Lemarchand; Victoria Lescano-Charreau; Julieta Ruiz; Nuria Caceres; Facundo Carrillo; Mariano Sigman; Joaquin Navajas

Abstract. Abstract Understanding when and why social interaction improves human judgment is a central question in the behavioural sciences. We examine whether collective accuracy improves when groups break complex estimation problems into simpler components and generate approximate intermediate estimates, a reasoning strategy we call Collective Fermi Estimation. Across three studies analysing 1140 online group deliberations in text chatrooms we study its causal effects and linguistic signatures. In Study 1 (N = 500), spontaneous use of problem decomposition predicts lower collective estimation error. Study 2 (N = 240) provides causal evidence: groups instructed to apply problem decomposition outperform groups instructed to combine their initial guesses. Study 3 (N = 160) shows that the benefits are larger when the strategy is applied collectively rather than individually. Altogether, we show that problem decomposition constitutes a key mechanism behind the wisdom of deliberative crowds and we provide tools to detect and promote it.

Read article →

Global trade-offs between consumption, carbon prices and equity

2026-08-05 · DOI 10.1038/S41467-026-75815-X

Mengyu Li; Jarmo S. Kikstra; Lorenz Keyer; Jason Hickel; Keywan Riahi; Manfred Lenzen

Abstract. As the remaining carbon budget continues to dwindle, economic growth strains efforts to limit global warming to 1.5 C. Reducing unnecessary production and consumption in high-income regions and among the wealthy, and organising production more around human needs and human well-being, may alleviate pressures on technology-driven decarbonisation strategies. We employ a modified integrated assessment model to explore the interaction between converging per-capita consumption, climate policy stringency and equity. Our pathways demonstrate that lower-consumption futures are associated with substantially lower warming, 0.3-1.1 C across different carbon price trajectories, relative to high-consumption futures. Comparatively, global convergence of incomes between regions has a small influence on global temperature outcomes (<0.1 C). These findings indicate that reducing high per-capita consumption in high-income regions is a primary lever for achieving futures with equitable and decent living standards while meeting global climate targets. While such transformations may face political challenges, our scenarios suggest that unconstrained growth futures make it considerably harder to remain within safe climate limits. The authors use a modified integrated assessment model to show that lower-consumption and more equitable futures could substantially reduce warming, highlighting the importance of reducing high consumption in affluent regions to meet climate targets.

Read article →

Proceedings of the National Academy of Sciences

Implicit biases change more slowly in larger cities

2026-08-06 · DOI 10.1073/PNAS.2514189123

Andrew J. Stier

The Santa Fe Institute

Abstract. Implicit biases remain a contributor to discrimination despite widespread endorsement of equality. Although there has been a large-scale reduction in sexuality, race, and skin tone biases in the past two decades, there is a current shortage of quantitative theories that describe how and why biases change. Here, I present a theory of implicit bias change, which focuses on changes in exposure due to cultural information and how the speed of these changes varies geographically. This model makes counterintuitive predictions when compared with previous findings, including 1) that biases change more slowly

in larger cities and 2) that one-off interventions are less effective in larger cities. I confirm these predictions using 14 million tests of implicit biases from U.S. residents over 14 y and experimental data from 1,150 individuals. Documenting the typical timescales and patterns that govern implicit bias change is an important step toward designing interventions that can facilitate further reductions in biases and prevent biases from increasing.
Read article →

Oxytocin increases trust in men low in dispositional trust

2026-08-06 · DOI 10.1073/PNAS.2602655123

Bodo Vogt; Paul Bengart; Carolyn Declerck; Ernst Fehr

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Abstract. In recent years, increasing skepticism regarding oxytocin's (OT) influence on social behavior arose. Low power, HARKing (hypothesizing after the results are known), and replication failures have clouded the field. Here, we directly address these concerns with a high-powered, preregistered study that offers robust evidence for a causal effect of OT on trust among males with a low disposition to trust. We recruited 359 low-trusting men who participated in a trust game with complete anonymity. Results show that OT administration significantly increased trusting behavior by roughly 15%, with consistent effects across regression models with and without controls for personality traits. A pooled data analysis incorporating a previous sample (n = 219) of low-trusting individuals further strengthens this conclusion, yielding a statistically significant 16.9% increase in trust. No interaction effect was found between OT and the degree of dispositional trust, suggesting OT's effect may be uniform across the low-trusting spectrum. These findings present a strong case for OT's selective trust-enhancing role. By isolating OT's impact within a well-defined subpopulation and experimental context, this study resolves a key question in the debate over neurobiological determinants of trust.
Read article →

Claims of sex-dependent effects in the behavioral and brain sciences are infrequently supported by valid statistical comparisons

2026-08-04 · DOI 10.1073/PNAS.2608703123

Madeline T. Olivier; Andrew W. Brown; Simon Chung; Colby J. Vorland; Donna L. Maney

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Abstract. Interest in precision medicine has accelerated efforts to identify sex-dependent responses to treatments and interventions. Those efforts have, both historically and today, been concentrated in fields related to the brain and behavior. Here, we show that even though articles claiming sex-dependent effects are far more prevalent in the neurosciences than in other fields, those claims are infrequently supported by appropriate statistical evidence. Of 200 recent articles in the behavioral and brain sciences with a claim of a sex- or gender-dependent effect in the title, only 24% supported the claim with appropriate evidence-that is, they compared the effect statistically across sex and reported results consistent with the claim. In 9%, an appropriate test was described but the results were incomplete or missing; another 9% contained results that directly contradicted the claim. In most articles (58%), the effect was not compared statistically across sex. Across six brain-related research areas (behavioral sciences, clinical neurology, neurosciences, psychiatry, psychology, and substance abuse), neurosciences had the lowest rate of appropriate evidence (18%) and psychology the highest (39%). Studies on nonhuman animals supported the claim significantly less frequently (15%) than studies on human participants (34%; P = 0.002). The use of appropriate analytical approaches was unrelated to journal rank or the citation impact of the article. We conclude that claims of sex- or gender-dependent effects in the behavioral and brain sciences are often not supported by appropriate evidence, revealing a widespread inferential gap that may undermine translation of sex-dependent findings into evidence-based treatments.
Read article →

Optimizing infectious disease mitigation under dynamic conditions

2026-08-03 · DOI 10.1073/PNAS.2527395123

Laura Muller; Fabio Sartori; Jonas Dehning; Maximilian F. Eggl; Viola Priesemann

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Abstract. Mitigation measures are essential for controlling the spread of infectious diseases during pandemics and epidemics, but they impose considerable societal, individual, and economic costs. We developed a general framework that combines simulation of disease dynamics with optimal control to determine mitigation strategies that balance infection and mitigation costs. Optimizing this trade-off, we identified three surprising effects: first, assuming a constant reproduction number R 0 , the optimal response is typically "all-or-nothing": depending on disease severity, either strict mitigation or none at all is optimal, with intermediate levels emerging only in restricted regimes that we characterize analytically. Second, under seasonal variations, optimal mitigation is stricter during winter. Interestingly, a single wave of infections still arises in spring, replacing the autumn/winter waves known for classical influenza. Third, during steady vaccination campaigns, even optimal mitigation can result in transient infection waves. Finally, we quantify the cost of delayed mitigation onset and show that even short delays can substantially increase total costs-if the disease is severe. Overall, our framework is easily applicable to general and complex settings and thereby presents a versatile tool to explore optimal mitigation strategies for endemic and pandemic infectious disease.

Read article →

Science Advances

Assessing pedestrian responses to autonomous and personal mobility robots in crowded public spaces

2026-08-07 · DOI 10.1126/SCIADV.AEF2576

Dominik Wojcikiewicz; Aude Billard; Diego Paez-Granados

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Abstract. Robots increasingly share spaces with people, supporting delivery services and mobility for the aging populations, yet their ability to share space comfortably lacks understanding and benchmarks for designers and policy-makers. We compared human-robot (HRI) and human-human (HHI) interactions across four real-world crowd datasets spanning Europe, North America, and Asia, using a unified pipeline to detect interactions, stratify by crowd density, and model pedestrian behavior. Local motion patterns (speed, acceleration, and jerk) remained closely matched between HRI and HHI across all densities. In contrast, proxemics diverged, with effects that grew approximately linearly with robot speed and weakened under higher crowding: In the dataset with the faster navigating robot, pedestrians maintained about 0.23 meters more clearance around the robot than around other pedestrians under less crowded conditions and about 0.05 meters more under more crowded conditions, while in the dataset with the predominantly stationary robot, the corresponding differences were small and inconsistent across crowding levels. The main conclusions were robust to parameter variations and remained stable across a broad range of motion-processing and interaction-labeling settings. Our findings provide density- and speed-aware benchmarks for proxemics in social robot navigation and empirically grounded targets for design, evaluation, and modeling across robotics and urban mobility.

Read article →

Escalation dynamics and the severity of wars

2026-08-07 · DOI 10.1126/SCIADV.AEB1705

Aaron Clauset; Barbara F. Walter; Lars-Erik Cederman; Kristian Skrede Gleditsch

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Abstract. Although very large wars remain an enduring threat in global politics, we lack a clear understanding of how some wars become large and costly, while most do not. There are three possibilities: Large conflicts begin with intense fighting, accumulate severity over a long duration, or escalate in intensity over time. Using detailed within-conflict data on civil and interstate wars (1946-2008), we show that escalation dynamics-variations in fighting intensity within an armed conflict-are the primary mechanism for producing large conflicts. Civil wars, however, tend to deescalate once they become very large, limiting their overall severity, while interstate wars exhibit a persistent risk of continual escalation. Simple models of within-conflict severity dynamics show that this distinction compactly explains the historical size distributions of civil and interstate wars, and unconditioned escalation is a plausible mechanism for Richardson's law-the power law pattern in the frequency and severity of interstate conflicts. The dynamics of escalation within conflicts have broad implications for conflict theory and risk assessment.

Read article →

Higher-order trade-offs in hypergraph community detection

2026-08-07 · DOI 10.1126/SCIADV.AEF2184

Jiaze Li; Michael T. Schaub; Leto Peel

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Abstract. Extending community detection from pairwise networks to hypergraphs introduces fundamental theoretical challenges. Hypergraphs exhibit structural heterogeneity with no direct graph analog: Hyperedges of varying orders can connect nodes across communities in diverse configurations. Crucially, this means that any hypergraph community detection algorithm must make a choice regarding which order or balance of hyperedges it prefers to maintain or split during partitioning. We address these challenges by developing a unified framework for community detection in nonuniform hypergraphs under the hypergraph stochastic block model. We introduce a general signal-to-noise ratio that enables a quantitative analysis of trade-offs unique to higher-order networks, such as which hypergedges we choose to split across communities and how we choose to split them. Building on this framework, we derive a Bethe Hessian operator for nonuniform hypergraphs that provides efficient spectral clustering with principled model selection. We characterize the resulting spectral detectability threshold and compare it to belief propagation limits, showing the methods coincide for uniform hypergraphs but diverge in nonuniform settings. Synthetic experiments confirm our analytical predictions and reveal systematic biases toward preserving higher-order and balanced shape hyperedges. Application to empirical data demonstrates the practical relevance of these higher-order detectability trade-offs in real-world systems.

Read article →

Steppe productivity decline contributed to smallpox outbreak and the collapse of the last nomadic empire in the 1750s

2026-08-07 · DOI 10.1126/SCIADV.AEE4285

Wanru Tang; Keyan Fang; Hans W. Linderholm; Etienne Boucher; Zepeng Mei; Jinbao Li; Shuheng Lin; Feifei Zhou; J. Camarero; Heikki Seppa; Fahu Chen

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Abstract. The collapse of the Dzungar Khanate in the 1750s marked the end of the world's last nomadic empire and reshaped Eurasian geopolitics. However, whether environmental and societal factors interplayed and contributed to its collapse remains unclear. Here, we reconstructed steppe productivity variability across the Dzungar heartland since 1625 using 441 tree-ring series based on an extreme growth ratio. Our record shows that the Khanate's collapse coincided with the lowest steppe productivity in the past four centuries during 1751-1761. The weakening societal resilience approaching a tipping point intensified the effects of the rapid steppe productivity decline, which cascaded into famine, conflict, and trade breakdown. Smallpox then reemerged and escalated into a catastrophic epidemic, transforming the environmental crisis into the collapse of the Dzungar Khanate. Our findings underscore the urgent need to strengthen pastoral resilience and epidemic surveillance in steppe regions as climate stress intensifies and cross-border epidemic threats escalate.
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Nature

How scientists can tackle the paradox of political polarization

2026-08-04 · DOI 10.1038/D41586-026-02425-4

Peter Calow

Abstract. Abstract unavailable.
Read article →

Science

How should India approach AI?

2026-08-06 · DOI 10.1126/SCIENCE.AEL0463

Akash Kapur; Arvind Narayanan

Akash Kapur is a visiting fellow at Princeton University and a senior fellow at New America, Princeton, NJ, USA.; Arvind Narayanan is a professor of computer science at Princeton University and the director of its Center for Information Technology Policy, Princeton, NJ, USA.

Abstract. In an unstable geopolitical environment, countries are looking to secure access to artificial intelligence (AI) capabilities, which are seen as essential to economic growth, scientific research, and national security. Few have placed sovereignty as squarely at the center of their AI strategy as India-and few are as well positioned to achieve it. But to do so, it needs to look beyond infrastructure-chips, data centers, and large language models-and pay more attention to the application layer above them: AI agents, the software and tools that accomplish real-world work. If the effort succeeds, India may pioneer

a template for genuine AI sovereignty that other nations can follow.
Read article →

The bridge between science and policy: How IPBES redefined global conservation policy and its path to societal impact

2026-08-06 · DOI 10.1126/SCIENCE.AEK0207

Patricia Miloslavich

Patricia Miloslavich is Program Lead of the Antarctic Monitoring Program at the Australian Antarctic Division, Department of Climate Change, Energy, the Environment, and Water, Kingston, Tasmania, Australia, and co-chair of the Monitoring Assessment of the Intergovernmental Science-Policy Platform for Biodiversity and Ecosystem Services (IPBES)

Abstract. Abstract unavailable.
Read article →

The climate benefits of retiring a fully operational internal combustion engine vehicle

2026-08-06 · DOI 10.1126/SCIENCE.ADV5441

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Abstract. Internal combustion engine (ICE) vehicles represent the single largest source of carbon dioxide emissions for most US households. Although battery electric vehicles (BEVs) considerably reduce use-phase emissions, the transition poses a fundamental life cycle optimization challenge: Is there a net climate benefit to retiring a functional ICE vehicle before its end of life? Retiring a functioning ICE vehicle incurs a fundamental trade-off between BEV manufacturing emissions and decreased vehicle operation emissions. In this study, we evaluated this trade-off across a comprehensive range of retirement scenarios and found that scrappage-and-replacement yields consistent emissions reductions across most contexts, including the retirement of new ICE vehicles. Although scrapping a functional asset remains economically prohibitive under current market conditions, these results demonstrate a major mitigation potential for policy interventions, such as enhanced scrappage subsidies, to address the emissions of an increasingly durable ICE fleet.
Read article →

Previous Week Catch-Up

Political Communication

Happiness in Political Communication by Men and Women Members of the US Congress

2026-07-29 · DOI 10.1080/10584609.2026.2705676

Constantine Boussalis; Travis G. Coan; Mirya R. Holman

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Abstract. Abstract unavailable.
Read article →

Networked Counterpublics and Contentious Publicity: A Cross-Platform Analysis of QAnon on Parler and Twitter

2026-07-27 · DOI 10.1080/10584609.2026.2708132

Sang Jung Kim; Jordan Foley; Yunkang Yang; Keyeun Lee

Seoul National University; Texas A&M University; University of Iowa; Washington State University

Abstract. Extremist online movements often leverage social media affordances to shape in-group identity and influence broader publics through agenda-setting practices. This study uses QAnon as a case of a networked counterpublic to examine how these processes unfolded between 2018 and 2021. Drawing on affordance theory, we analyze how QAnon discourse activated the affordances of visibility, association, and persistence on Parler and Twitter - two platforms similar in technical features but distinct in their cultures of use. We identify differentiated affordance-activation practices that support different counterpublic functions: for example, hashtag stacking on Parler can be interpreted as contributing to identity-oriented meaning-making, whereas calls to "do your own research" on Twitter may function as epistemic invitations aimed at drawing mainstream users into conspiratorial interpretation. We further show that QAnon's contentious publicness - the contestation of pro- and anti-QAnon claims - shifted dynamically across the movement's developmental stages, and that identical informational resources (shared URLs) generated divergent discursive trajectories across platforms. By tracing these temporal and cross-platform dynamics, we illuminate how extremist counterpublics navigate visibility, identity work, and public contestation, offering insights for the study of political communication in an era of networked extremism.

Read article →

Communication Research

The Bio-Functional Audience: Shared Physiological Alignment to Narrative Media

2026-07-28 · DOI 10.1177/00936502261472026

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Abstract. Communication research into audience responses has traditionally emphasized how individual or group differences create variance in interpretation while overlooking a more fundamental phenomenon: the capacity of the same message to induce similar responses across receivers. We introduce a bio-functional definition of the audience, a temporal collective bound together by the moment-to-moment alignment of its members' physiological responses. In a large-scale experiment (N = 198) monitoring eye-tracking, heart rate, and electrodermal activity during a film, we mapped a processing hierarchy in which audience response alignment was strongest for gaze but also extended into downstream autonomic responses. Furthermore, a temporal manipulation of the narrative sequence abolished aligned responses, confirming that the induced audience state is driven by the message content. We conclude that messages function as alignment devices that reduce individual noise and bring distinct individuals into a shared processing state, thus providing a grounded, signal-based definition of the audience as a bio-functional entity.

Read article →

Political Analysis

The Power of Prognosis: Improving Covariate Balance Tests with Outcome Information

2026-07-31 · DOI 10.1017/PAN.2026.10050

Clara Bicalho; Adam Bouyamourn; Thad Dunning

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Abstract. Abstract Scholars frequently use covariate balance tests to test the validity of natural experiments and related designs. Unfortunately, when measured covariates are unrelated to potential outcomes, balance is uninformative about key identification conditions. We show that balance tests can then lead to erroneous conclusions. To build stronger tests, researchers should identify covariates that are jointly predictive of potential outcomes; formally measure and report covariate prognosis; and prioritize the most individually informative variables in tests. Building on prior research on "prognostic scores," we develop bootstrap balance tests that upweight covariates associated with the outcome. We adapt this approach for regression-discontinuity designs and use simulations to compare weighting methods based on linear regression and more flexible methods, including machine learning. The results show how prognosis weighting can avoid both false negatives and false positives. To illustrate key points, we study empirical examples from a sample of published studies, including an important debate over close elections.
Read article →

American Political Science Review

A Democratic Theory of Labor Unions

2026-07-30 · DOI 10.1017/S0003055426101804

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King's College London

Abstract. Recently, political theorists have been increasingly concerned with the implications of democratic values for work. This article contributes to these debates by examining the main vehicle through which workers, historically, have sought democracy at work: the labor union. The article defends a vision of economic democracy that arises from the practice of labor unions. It then emphasizes the distinctive features of labor unions that make them potential vehicles for democracy in other domains. Labor unions are particularly important means of advancing democratic values for three reasons: (1) their ability to appeal to the short- and mid-term self-interest of their members; (2) their anchor in the material reproduction of society; and (3) that they rely on attitudes of cooperation and solidarity among their members. Yet these three features also point to potential limitations of labor unions as laboratories of democracy.
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Redistricting Reforms Reduce Gerrymandering by Constraining Partisan Actors

2026-07-30 · DOI 10.1017/S0003055426101610

CORY McCARTAN; CHRISTOPHER T. KENNY; TYLER SIMKO; EMMA EBOWE; MICHAEL Y. ZHAO; KOSUKE IMAI

College of William and Mary; Harvard College; Harvard University; Pennsylvania State University; Princeton University; University of Michigan

Abstract. Political actors often manipulate redistricting plans to gain electoral advantages, a process known as gerrymandering. Several states have implemented institutional reforms to address this problem, such as establishing map-drawing commissions. Estimating the impact of these reforms is challenging because each state structures its processes and rules differently. We model redistricting as a sequential game whose equilibrium solution summarizes multistep institutional interactions as a univariate score. We argue that this score measures the leeway political actors have over the partisan lean of the final plan. Our differences-in-differences analysis demonstrates that reforms reduce partisan bias and increase competitiveness by constraining partisan actors. We perform a counterfactual policy analysis to estimate the effects of enacting recent reforms nationwide. Though commissions are likely to reduce bias, reforms that restrict partisan actors in multiple ways, such as removing veto points (Michigan), are more effective than commissions where parties retain some control (Ohio).
Read article →

Contested Governance and Civilian Prosocial Behavior: Evidence from Lab-in-the-Field Experiments in India

2026-07-29 · DOI 10.1017/S0003055426101701

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Abstract. How does contested governance by states and competing armed actors affect the behavior of the governed? During civil conflicts, rebel organizations frequently provide public goods, judicial arbitration, and security, but these forms of governance often operate alongside local state institutions. Beyond avoiding violence, therefore, noncombatants must navigate the daily uncertainty of local governance by states and armed groups vying for territorial control. This, we argue, shapes their quotidian interactions with one another. Using a series of lab-in-the-field experiments and surveys from Naxal-affected Bihar, India, we find that individuals living under such contested governance exhibit greater interpersonal trust and trustworthiness and less opportunistic behavior. We propose two mechanisms to explain these patterns-fearing intra-community retaliation or substituting for the state with interpersonal reliance. Most evidence is only consistent with the latter. By connecting governance, rebellion, and social capital, this work makes important contributions toward understanding the micro-politics of local order during longstanding armed conflicts.

Read article →

Is Blood Thicker Than Water(-and-Earth)? Partisanship Geography in Imperial China and the Limits of Quantitative History

2026-07-29 · DOI 10.1017/S0003055425100853

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University of Detroit Mercy

Abstract. Using Wang Anshi's Reform as a case study, Yuhua Wang's "Blood is Thicker Than Water" (American Political Science Review , 2022) argues that geographically dispersed kinship networks incentivize political elites to support the building of a strong state, whereas geographically concentrated kinship networks undermine state building. In this article, I improve the sample with the help of additional historical records and show that the reported correlation fails to replicate when either the control issue or the outlier issue is considered. I then propose two alternative hypotheses on the role of kinship network dispersal in Wang Anshi's Reform, based on two common explanations of its partisanship geography in Chinese historical scholarship, that is, status competition and policy adaptiveness. After examining those hypotheses respectively, I reflect on the limits of quantitative history that this case illustrates.

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Still Instrumentally Inclusive

2026-07-28 · DOI 10.1017/S0003055426101828

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University of Southampton; Vrije Universiteit Amsterdam

Abstract. Do individuals in Western democracies shift their views on LGBT+ inclusion when exposed to opposition from Muslim out-groups? And is this "homonationalist" responsiveness stronger among anti-immigration individuals? Our article "Instrumentally inclusive [...]" addressed these questions using representative data from Britain (Study 1) and crowd-sourced data from Spain (Study 2). We revisit Study 2 in response to concerns about post-stratification weighting a convenience sample and erroneously inconsistent use of robust standard errors. Varying population reference parameters, weight caps, variance estimators, subgroup definitions, and covariate adjustments, we find strong support for the original findings. Neither reasonable alternative weighting schemes nor corrected variance estimators substantively change the results. In fact, the originally published point estimates are equal to or smaller than the median across alternative specifications. Consistent with

the Article, we find that treatment increases support for LGBT+ inclusion in Spain, without consistent identifiable variation by immigration attitudes.
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Nature Communications

Foraging models explain human exploration in uncertain tasks

2026-07-31 · DOI 10.1038/S41467-026-75773-4

Meriam Zid; Veldon-James Laurie; Jorge Ramirez-Ruiz; Alix Lavigne-Champagne; Akram Shourkeshti; Dameon C. Harrell; Alexander B. Herman; R. Becket Ebitz

Abstract. Abstract Different fields have fundamentally different models of how decisions are made. Psychology and neuroscience tend to assume that decisions are made by calculating and comparing the values of all options. In ethology, conversely, decisions tend to be binary regardless of the number of options: Decision-makers calculate the value of continuing to exploit some option and explore only when this value drops below a threshold. Because these fields use incompatible methods, it remains unclear which view better describes human decision-making. We find that humans use compare-to-threshold computations even in classic compare-alternative tasks. Because the reinforcement-learning models typically used in the cognitive and brain sciences depend on compare-alternative computations, we also develop a compare-to-threshold foraging model. Compared to previous models, the foraging model better fits participant behavior, predicts the tendency to repeat choices, and predicts held-out participants that were almost impossible under traditional compare-alternative models. These results suggest that humans use compare-to-threshold computations in more environments than were previously known.
Read article →

Life course trajectory segregation and political polarization in contemporary Sweden

2026-07-30 · DOI 10.1038/S41467-026-75072-Y

Bo Malmberg; Juta Kawalerowicz; Eva K. Andersson

Abstract. Abstract Understanding how and why places differ geographically has long been central to human geography-a question sharpened by the growing geographical polarization of political views. Spatial polarization can result from an uneven geographical distribution of life-course trajectories, trajectories that emerge from individuals' successive positions across interconnected, possibly spatially differentiated social fields such as education, employment, family, and housing. Here we show that the uneven geographical distribution of such trajectories produces landscapes where contrasting life experiences, opportunities, and aspirations become inscribed in geographical space. A key finding is that segregation intensifies over the life-course, with young adults substantially less segregated than older adults, particularly after homeownership transitions. Moreover, studies relying on single indicators such as income and education may underestimate residential segregation strength. Homeownership is a major segregation driver, creating highly separated residential areas in later life phases. The spatial concentration of trajectory types also corresponds with geographical patterns in political polarization.
Read article →

Low-burden AI approach for cross-national early identification of cognitive impairment using real-world questionnaire response behaviours

2026-07-28 · DOI 10.1038/S41467-026-76071-9

Hongxin Gao; Stefan Schneider; Jenny Harris; Raymond Hernandez; Doerte U. Junghaenel; Arie Kapteyn; Pey-Jiuan Lee; Arthur Stone; Elizabeth Zelinski; Danny Maupin; Bart Orriens; Haomiao Jin

Abstract. Abstract Under-recognition of cognitive impairment remains a major global public health challenge, particularly in low- and middle-income countries. Many cases go unrecognised because conventional cognitive assessment is

resource-intensive and difficult to scale across diverse populations. Here we show that easily captured indicators of reduced survey response quality, derived from how older adults answer routine psychosocial questionnaires, can identify cognitive impairment risk across countries. Using data from 45,604 older adults across five population-based ageing cohorts, we develop a cross-nationally generalisable machine-learning pipeline based on a tabular foundation model. Especially, the model jointly trained across the United States, England, India and Mexico outperforms country-specific models, achieving up to a 0.12 absolute gain in the area under the receiver operating characteristic curve (approximately 20% relative) in an external validation cohort from China. We also identify a cross-national implicit feature transfer phenomenon, in which cohorts lacking certain predictors benefit from joint training with cohorts in which these predictors are present. Decision-curve analyses indicate consistent potential public health benefit across countries by supporting population-level prioritisation of higher-risk subgroups. This low-burden approach offers an equitable and scalable strategy for generating population- and community-level evidence on cognitive impairment risk in culturally diverse and resource-constrained settings.
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Proceedings of the National Academy of Sciences

Social dynamics of AI adoption in parents' educational decisions

2026-07-31 · DOI 10.1073/PNAS.2533193123

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Abstract. When adopting a performance-enhancing technology offers short-run relative advantages, competitive pressure can drive diffusion even if most adopters would prefer a collective ban. We study this dynamic in the context of unrestricted AI tools for teenagers' education. In preregistered, incentive-compatible experiments with approximately 2,000 parents in the United States, Canada, and the United Kingdom, we elicit willingness to pay for a three-month premium subscription to a leading general-purpose AI platform. Demand increases steeply with the share of other teenagers using AI: Moving perceived peer take-up from 20% to 80% raises willingness to pay by more than 60%. We then test whether information about emerging evidence on potential downstream skill effects changes behavior. Risk information substantially shifts incentivized beliefs about longer-run impacts but has little effect on individual demand, yet it strongly increases support for school-level restrictions. Follow-up experiments show that peer-adoption information does not significantly shift perceived AI quality, the importance parents assign to core skills, or support for bans, pointing to social pressure rather than social learning as the primary channel. The results reveal a rat-race dynamic in which individual adoption responds to peers even as collective preferences shift toward restriction.
Read article →

A demographic theory of similarity-biased social learning

2026-07-30 · DOI 10.1073/PNAS.2606741123

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Abstract. We develop a demographic theory of similarity-biased social learning that formalizes our understanding of when and why individuals should preferentially copy others that look or act like them. We build an evolutionary model in which individuals can either learn on their own or copy others from a demonstrator pool that contains varying proportions of in-group and out-group members, and where group tags can be more or less informative about local knowledge. We find that where social learning becomes common, selection favors copying biases that track the direction of informational advantage-toward the group that tends to be better adapted to local conditions, including an antisimilarity bias when tags are negatively associated with local correctness (as may be the case for some immigrant communities). We also find conditions in which a

similarity bias can stabilize social learning when such learners are already common, but not when they are rare, with implications for the role of group identities in cultural evolution. We discuss implications for understanding parochialism as risk aversion, majority-minority dynamics, the sociology of immigration, and the lasting impacts of colonialism.
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Parenthood decisions in uncertain times: Experimental evidence from four countries

2026-07-28 · DOI 10.1073/PNAS.2601690123

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Abstract. Fertility rates are declining across much of the globe, yet the forces driving this decline have become increasingly complex, and classic theories have become less adequate to explain recent trends. While economic and gender-related explanations have long dominated fertility research, emerging global uncertainties, such as climate change, democratic backsliding, economic crises, and regional conflicts, may also shape how individuals evaluate childbearing. In addition to the more classical factors, this cross-national study examines how macrolevel uncertainties influence perceptions of fertility intentions in Italy, Argentina, the United States, and Germany. These countries offer distinct fertility trajectories and face unique combinations of sociopolitical and environmental challenges. Using a conjoint experiment, we examine how individuals evaluate, across hypothetical scenarios, their intention of having a child under varying household, policy, and macrolevel conditions. Across all four countries, material security remains central: household income is the strongest predictor of fertility intentions, with homeownership and childcare playing smaller roles. At the same time, perceptions of macrolevel stability, including expectations about climate conditions, economic prospects, democratic functioning, and peace, exert substantial and meaningful effects, comparable to homeownership and childcare provision but more pronounced than those associated with gender-role arrangements. Cross-country differences show that macroeconomic stability and climate change are most consequential in Argentina, while in the United States respondents are less responsive to geopolitical conflict. These findings show that individuals evaluate childbearing not only through their immediate circumstances, but also through expectations about broader societal futures, highlighting the importance of incorporating macrolevel uncertainty when explaining contemporary fertility decline.
Read article →

Psychological inoculation against climate doom

2026-07-28 · DOI 10.1073/PNAS.2609074123

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Abstract. Individuals encounter "climate doom," extreme portrayals of climate change, on social media. While climate doom may be intended to motivate action, it may also be perceived as manipulative or elicit hopelessness. Yet, no research has examined whether psychological inoculation helps to improve recognition of climate doom or influences engagement with it on social media. In this preregistered study (n = 521), we tested whether inoculation improves discernment of climate doom and influences engagement with it by collecting behavioral data using a social media simulator. We find that inoculation substantially improves discernment of climate doom and that this is due to increased perceived manipulation of doom messaging specifically (d = 0.68). There was no interaction between the inoculation condition and political leaning and while higher perceived manipulation of doom content predicted lower behavioral intentions, inoculation did not moderate this relationship. Inoculation increased engagement, particularly disliking and flagging of doom content, and decreased sharing. Overall, our results should lead climate communicators to reconsider doom messaging as a mobilization strategy and demonstrate that inoculation is an effective tool for increasing recognition of, and critical engagement with, climate doom

across the political spectrum.
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ICE arrests, 2015-2026: Variation in targeting, method, and geography

2026-07-27 · DOI 10.1073/PNAS.2618096123

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Abstract. We analyze and decompose administrative data on all 1.6 million US Immigration and Customs Enforcement (ICE) arrests from October 2015 to March 2026. Our results reveal that the reality of immigration enforcement diverges sharply from the public narrative that ICE arrests are necessary to protect public safety by removing people who commit crimes: Although arrests spiked at the outset of both Trump presidencies, the share of arrested individuals with criminal convictions fell significantly, with especially marked declines in 2025. A shift in ICE tactics partly explains this pattern, but even conditional on tactic, the share with a criminal conviction declined as arrests rose. Moreover, we find substantial geographic heterogeneity in ICE methods, despite nearly universal declines in the criminal conviction rate. Our findings provide important evidence for policymakers, the general public, and researchers studying this period of immigration policy.
Read article →

Science Advances

Most people who agree with you won't protest: Predictors of marches and demonstrations from six decades of research

2026-07-31 · DOI 10.1126/SCIADV.AEA8409

Shelley Boulianne; Jennifer Earl

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Abstract. Protest and activism are endemic and potent features of many nations' political life and critical potential counters to democratic backsliding and authoritarianism. In this review article, we meta-analytically examine six decades of research findings on what explains participation, focusing particularly on two well-studied protest forms: marches and demonstrations. Focusing on these iconic and common forms of protest participation ensures accurate research comparison, allowing us to identify sets of variables that are consistently important, others that are not consistently important, and still others that need additional research. Our focus on survey-based studies also allows us to compare research that includes both participants and nonparticipants, which is critical for methodologically sound comparisons. Our analysis of 181 survey-based studies (with more than 1800 different estimates) shows that survey research strongly supports the relevance of political interest, political ideology, membership in organizations, and digital/social media use in explaining participation, as well as demographic variables such as education, age, and living in an urban center. Our review also offers important suggestions for future survey research and micromobilization research more broadly.
Read article →

Untrustworthy sources on Facebook and Instagram in 2020: Concentrated exposure but no attitudinal effects

2026-07-31 · DOI 10.1126/SCIADV.ADZ6502

Olivier Bergeron-Boutin; Brendan Nyhan; Jaime Settle; Emily Thorson; Magdalena Wojcieszak; Taylor Brown; Adriana Crespo-Tenorio; Carlos Velasco Rivera; Hunt Allcott; Pablo Barbera; Drew Dimmery; Deen Freelon; Matthew Gentzkow; Sandra Gonzalez-Bailon; Andrew M. Guess; Edward Kennedy; Young Mie Kim; David Lazer; Neil Malhotra; Devra Moehler; Jennifer Pan; Daniel Robert Thomas; Rebekah Tromble; Arjun Wilkins; Beixian Xiong; Chad Kiewiet de Jonge; Annie Franco; Winter Mason; Natalie Jomini Stroud; Joshua A. Tucker

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Abstract. Despite concern about exposure to content from untrustworthy sources on social media, little is known about the frequency or effects of exposure to their content. We examine 2020 data from all active US adults on Facebook and Instagram to measure exposure to content from Pages, groups, and web domains on Facebook and public accounts on Instagram that repeatedly publish misinformation. We find that average users saw relatively little content in their feeds from these untrustworthy sources; exposure was highly concentrated. A multimonth field experiment during the 2020 election among consenting users reduced feed-based exposure to content from untrustworthy sources by approximately 70% on both platforms but had no measurable effects on numerous preregistered outcomes, even among participants with high pretreatment exposure. Our results demonstrate that a feasible platform intervention can successfully reduce exposure to content from untrustworthy sources but suggest that these changes are unlikely to have immediate effects on attitudes and beliefs.

Read article →

Nature

Relatively few Black people benefited in the 1930s and 1940s from mortgages backed by the US government

2026-07-29 · DOI 10.1038/D41586-026-02246-5

Abstract. Newly digitized mortgage records from 38 US states reveal the profound racial inequality of two mortgage programmes during the 1930s and 1940s. These loans, supported by the Federal Housing Administration and the Veterans Administration, were mainly responsible for the emergence of a US society that idealized home ownership. However, Black households were systematically excluded from receiving such loans. By contrast, immigrants received mortgages proportional to their number. Immigrants, who were mainly from Europe at the time, were proportionally represented among loan recipients in the United States.

Read article →

Science

When the future loses its voice

2026-07-30 · DOI 10.1126/SCIENCE.AEJ7915

Anne Goujon

Anne Goujon is director of the Population and Just Societies Program at the International Institute for Applied Systems Analysis, Laxenburg, Austria, and leader of the Population Data and Projections research group at the Vienna Institute of Demography of the Austrian Academy of Sciences, Vienna, Austria.

Abstract. Across many high-income societies, anxiety about demographic decline has become commonplace. Fertility rates have fallen below replacement level, populations are aging rapidly, and governments worry about labor shortages, pension sustainability, and the future of welfare systems. These concerns are legitimate. Yet the conventional narrative does not always capture another important consequence of declining birth rates: dejuvenation, the progressive reduction of the demographic, economic, cultural and political weight of younger generations in society. Although low fertility alone does not immediately transform institutions, the cumulative effect of decades of low births combined with rising longevity does. The result is a thinner presence of younger people in the labor market, in entrepreneurship and innovation, and in the democratic debates in which long-term priorities are contested.

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Privacy-preserving tools banned in census

2026-07-30 · DOI 10.1126/SCIENCE.AEL0161

Jeffrey Mervis

Abstract. Critics say goal is to reduce immigration and add Republican seats in Congress

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Rigor is not readiness: The PERL framework for evidence-based policy

2026-07-30 · DOI 10.1126/SCIENCE.AEF3529

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Abstract. Policy Evidence Readiness Levels help decision-makers assess evidence maturity and uncertainty

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